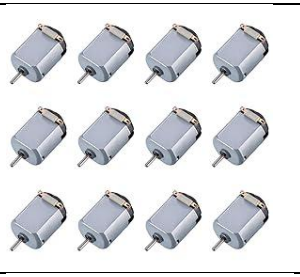
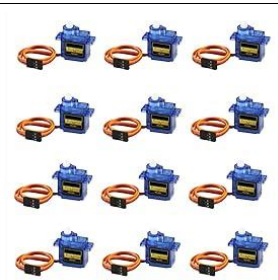
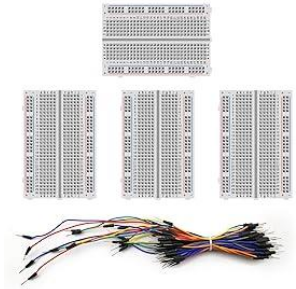
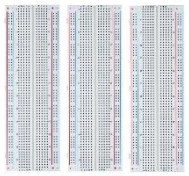
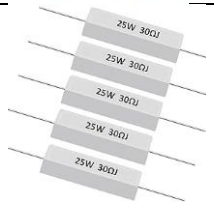
Parts		Spec	Links
Aluminum Rod (precision ground)		1/4in diameter stock for bearing shafts (precision ground – avoid use except for bearing mounts)	https://www.mcmaster.com/9062K26/
Aluminum Rod (raw)		1/4in diameter stock	https://www.mcmaster.com/4634T32/
spring			https://www.mcmaster.com/9657K414/
Bearings		1/4in ID 3/4in OD, 9/32in depth	https://www.amazon.com/dp/B0DX73ST8V?ref=ppx_yo2ov_dt_b_fed_asin_title
Shims for 1/4in shaft		1/4in ID 0.032 thick,3/8in OD	https://www.mcmaster.com/91124A160/

DC motor			https://www.amazon.com/Topoox-15000RPM-Electric-Science-Projects/dp/B07JYM8H18/ref=sr_1_7?keywords=dc+motor&qid=1578352696&sr=8-7
Servomotor			https://www.amazon.com/gp/product/B08FJ27Q1H/ref=ewc_pr_img_2?smid=A3CX4TQNUXMB0L&psc=1
DC gear motor			https://www.amazon.com/AEDIKO-Motor-Gearbox-200RPM-Ratio/dp/B09N6NXP4H?crid=29EY6GZ2FN48Z&dib=eyJ2IjoiMSJ9.JGma5iFHJVwHTdPfEkWG2S9xZvlwhYDV9IlPYDOdV46Ig9YjfBr3oFCBZmqbGLSfRIeF1_a8W7Mk6NL1IJbhsPDjW2XWDDmAbhPA6gkIx4ZQ8g_AJMjCub_UmEpZaBOixeFLcN6GFAD4yljpcC48JTpQkjy8BXzZLxyQkxKXAE8_Yw457GiGgZXHPhSZmBBg_KXEWyb8AH8sYpYR0VK5CeISRo68fAsWX9ASy6K6GgRx-gupaaCgFoQMGBdyyx5JnazNI7AB3QPTHVCNZLYCm26ix550WeKt2WwUMZ3U1mM.H9Eqnp-4xfZmdjsnLGZa87ddUTtLvYt4uE6blMbOIsY&dib_tag=se&keywords=dc+gearmotor&qid=1768001246&sprefix=dc+gearmoto%2Caps%2C201&sr=8-17

Solenoid		12V	https://www.amazon.com/JF-0530B-Electromagnet-Keepping-Travel-Solenoid/dp/B0F1KC3GCK?crid=2NX5NWRBPPGUS&dib=eyJ2IjoiMSJ9.lgVBc2T04wb6eNLKV6JFFEUNVKBeSufVyNvn9XcjLGLQQV98TDsCJ-yc3fF2KXc_SZEumoJhkb5hNKBFxP2gNbn_i6xTW05HCk3YoyOtRL30aHAEW8MpDmBV4H7pt89dMRhykAN6n6-zIOQkVhqDVqVqraAF9KvQW3RaCF6vy2P2yVe3q7TzIvXhE5aQJoMwQzmlGl6CNlVVRayv4ZO-CuE_-zeq-dUncNyVUNfFPfRSAt1rkDGjS-ZEZ5dhCk9_YKkjp-xejywda0-V_BtJzIAklqmtlH3FrCUfcfEzmTE.yYLpOuM7Dt_CqX48BRkxsdf7POfHEWrkFe2GZkvPTLY&dib_tag=se&keywords=JF-0530&qid=1768001317&sprefix=jf-0530%2Caps%2C167&sr=8-3&th=1
Small breadboard			https://www.amazon.com/gp/product/B07NVWR495/ref=sw_img_1?smid=A5JFCS1U77T7W&psc=1
big breadboard			
Power resistor			https://www.amazon.com/smseace-resistors-Tolerance-Computer-Apparatus/dp/B0B686MY4F/134-3586713-7764916?pd_rd_w=C7xlJ&content-id=amzn1.sym.751acc83-5c05-42d0-a15e-303622651e1e&pf_rd_p=751acc83-5c05-42d0-a15e-303622651e1e&pf_rd_r=Q8QWMWEQ5DGSH0ZSD8P0&pd_rd_wg=C1MX1&pd_rd_r=eaeca83e-8977-

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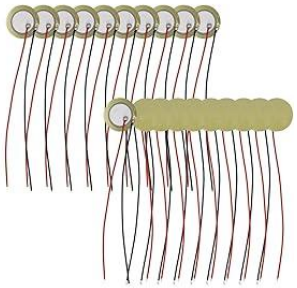
Power capacitor			https://www.amazon.com/gp/product/B09TVNDCJP/ref=ewc_pr_img_7?smid=A13KOVEB7K7KUP&th=1
piezo sensor			https://www.amazon.com/uxcell-Acoustic-Transducer-Microphone-Trigger/dp/B07RL6L1CF?crid=3MGQANTMQYO5Q&dib=eyJ2IjoiMSJ9.DhSQ848RBtu1c-xJ-MHmZQ7zdVk37f-v9W-CJ71t65fXH_oyarEjD6Lyoscx3OH8WiXFGddNTnbEXPK2JR1AAVQem_kU-2KsJ0NjSskgZHgo2BsOpDxYa3RxvEtPp2M-JX1LK2OVvJSqIOFeSiO5z5Ix5ZKIheBwCV9thr8PKrcHsuv7CREm1artQIC5lhH1IplfhYzjg7MuUAr95blLVbBNz2CXRbF1Ii0U9Mme0Z_DSfUdzksJDaOTr8kgxFZTyCBnQC8g-sBW74e8mBxR1qOlJ94NUd1-niajaRNTH2qzQyznAcRHYGZ4W7nHQWMNNee9v5NHLxv2suIuWs3ZdlNHJXJZyW8mA2FczHKN26dboHgwC0-siWw5V2eJaA8FajxxL2687S69_Up8-cULJ9hfkJ9qBb8Amz84x8QjkwETwicI6oU5hco6albwbOHw.NgdiscSu4DUXkYdNCCBI_nWPtXW0L762gP6OiMnfYSQ&dib_tag=se&keywords=piezoelectric+discs&qid=1735972877&sprefix=piezoelectric+di%2Caps%2C324&sr=8-13
Arcade button			https://www.amazon.com/uxcell-Mounting-Momentary-Button-Switch/dp/B07YJXR4N5?crid=T323L8KN9RI5&dib=eyJ2IjoiMSJ9.VDq2zyBXkb39OcrLu7CY1U_kdiGn2P8WzJ0B8swptGNK9_Cl1s77If7AW82RVLRTgrJftka5a5TOZuffaCq0-rSTBZ8_hHpHGoSVm5ofnlX_9IKwQw3AEMjpvm-KoL_3M_n5nJYowKyLCNn5utSWRwc5UxpETmSlcsMtgpfYYH_kqJRFEOd-yOZHUoDuytOxMapoKL5sn5jz1D-oR-IDyZXbHreSJnk3T2tp9ggu0T147jDoH_RSskk0MeO-9hfr5YYX99vzO_AzfAQKG3QW4gAfzhntWA38VMyB6Syw6XFW8aUZfH65sSc6IZnX9Z2T-qegj-lWQNL-fw565hIkx8XEb5ZS6R3evn9xu4-WOWTIIk0KM8GhCW4fcHJTEv1AO9D8qhW5G4SlFCXqrtB7uMnoBrHGBR1DnS8mnoCEfFsqSilLXcyaGIj5B

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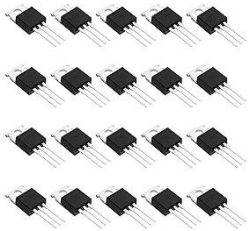
Arduino Mega			https://www.amazon.com/gp/product/B01H4ZLZLQ/ref=ox_sc_act_title_1?smid=A2WWHQ25ENKVJ1&psc=1
7 segment display		common cathode	https://www.amazon.com/gp/product/B07GTR8KB3/ref=sw_img_1?smid=A1THAZDOWP300U&psc=1
N-MOSFET (Power mosfet)			https://www.amazon.com/gp/product/B07MW1N4Q5/ref=sw_img_1?smid=A376OCJGR11S0Z&psc=1
Shift Register			https://www.amazon.com/gp/product/B0CRKZV6B9/ref=sw_img_1?smid=A37QXIKGEOBCOK&psc=1
IR LED			https://www.amazon.com/gp/product/B071VHB42J/ref=sw_img_1?smid=A2E7YALNIES9BY&psc=1

IR phototransistor			https://www.amazon.com/gp/product/B071VHB42J/ref=sw_img_1?smid=A2E7YALNIES9BY&psc=1
Ball (0.5” diameter bearing ball)			https://www.amazon.com/gp/product/B07L8N3CHG/ref=sw_img_1?smid=A23TVO4Z9GSJEQ&psc=1
potentiometers			https://www.amazon.com/HiLetgo-Single-Joint-Potentiometer-Variable-Resistors/dp/B07VJYH29F?dib=eyJ2IjoiMSJ9.9NYYa1419H7mSVLiKiz_V7Q9n3clnQ0GkUTKkwgRn5wfr8iPffGnHg73cn_OGu6dXPgjqzomSix_GfAeJMfzCYyhsy07s75mr6vO6sqxiSihjif3QEOBWdYa_CpyVfrFEciu3xvMiCx2ah5Zf8fAtbgjF3s5BbQlZTC9UHk4Lch9RRSrvnTjrsPgAjy2ux3bA03m_-uMxXgsNZm4_KipTc9jAVrjGLm_eRQhRKlBKO0.Mk9Mgjks0vc8bu_PIIr0hgLeoNqAghhK_RViWWszmGQ&dib_tag=se&keywords=potentiometer&qid=1735971608&sr=8-1
wires			https://www.amazon.com/Breadboard-Minidodoca-Magnetic-Assortment-Alligator/dp/B0DJ73V9ZJ?crid=CTTAW774SXV8&dib=eyJ2IjoiMSJ9.1JTtZYzqh1JVSNxn_zOlNKR-NWs8wozhUHSlzQX8IqcsNAuIYYeyIzhwHvJkpWEHUV_HWIvJhl9vZfJZzb0yR2Ptlt6IRrkdCRBw7an5u6UbENmoMDGdQOviSETHetl0oXVzGJRv3v9OOCPkdOkhkZva-XDkgRGRFj2davfOODcDgqCumeWtlhfEIm9ufX0xULEk6CHAj4K2iarkPYsoGLLHvs1rFBdIs8KlHjROuiA.LXLJ-A07Kmf16sMlPDFKA8i2Mc7uRkxh38LusMvb13Y&dib_tag=se&keywords=jumper%2Bwire%2Bkit&qid=1735971749&sprefix=jumper%2Bwire%2Bkit%2Caps%2C244&sr=8-5&th=1

speaker			https://www.amazon.com/Uxcell-a15080600ux0275-Internal-Magnet-Speaker/dp/B0177ABRQ6?dib=eyJ2IjoiMSJ9.MOZl22bWPSPHRfSY1ps0VdKks-EH_IddvvbrS5lP7wu2rqYJTyF_xv0opgiT_cl3P6RpempKNKOxLeeIifIc4tFpOeKbs_uNlKMOD9WhVucs8cvCiAeDeqQQjb1LRiwlo8or-SSz7EUpE71B1cZZfla9XQ6MjisztgA7YzkAPZU-I3-tAz2QZHnVv79tsd1-Hhg-5_u92vSEutYXgZ5DmzhfNJ4ByBaIU5wJbAbP9OI.7Gr-URt4HgIiRK7tFM-vUae0-VO-a4siGLKw7CXv-i8&dib_tag=se&keywords=speaker+piezo&qid=1735972943&sr=8-6